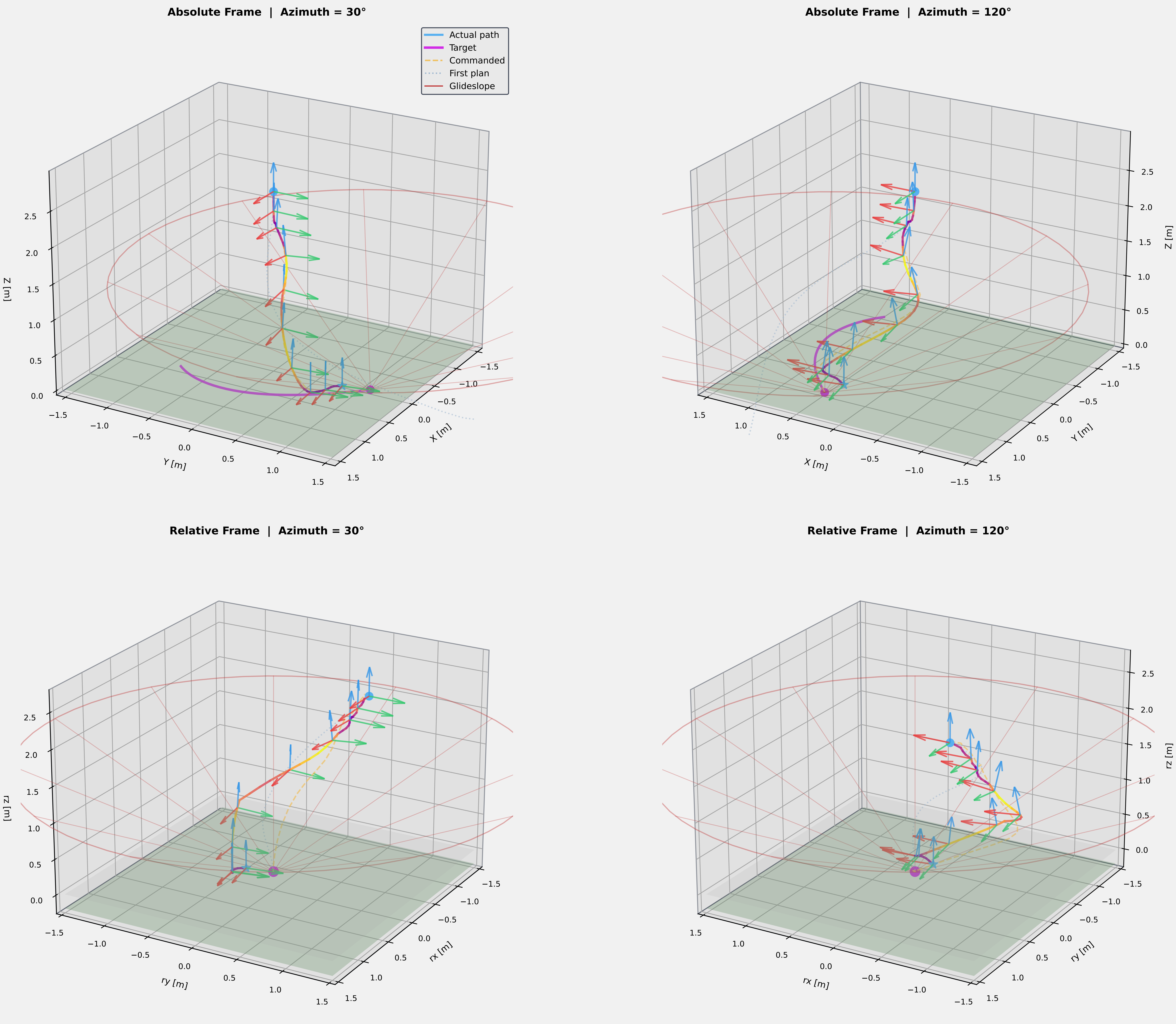
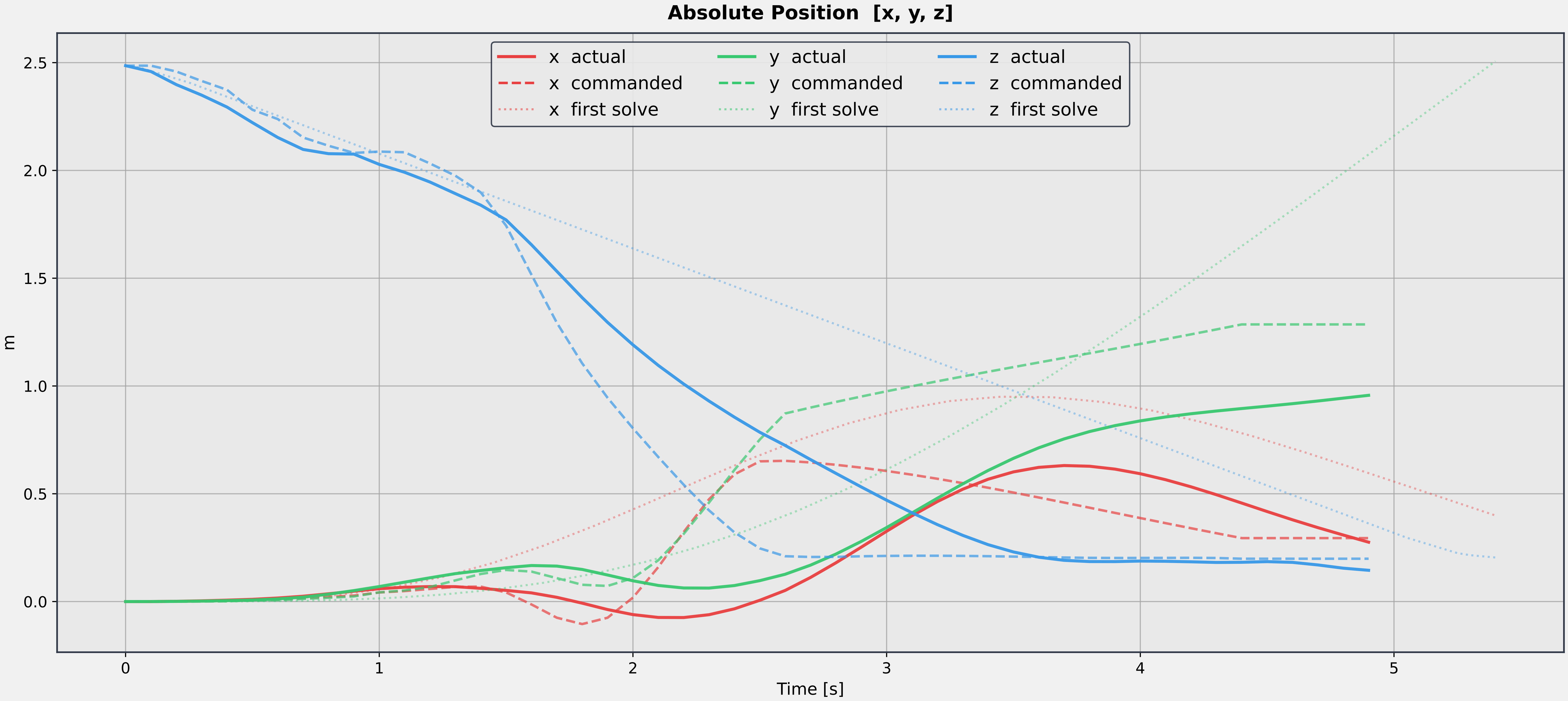
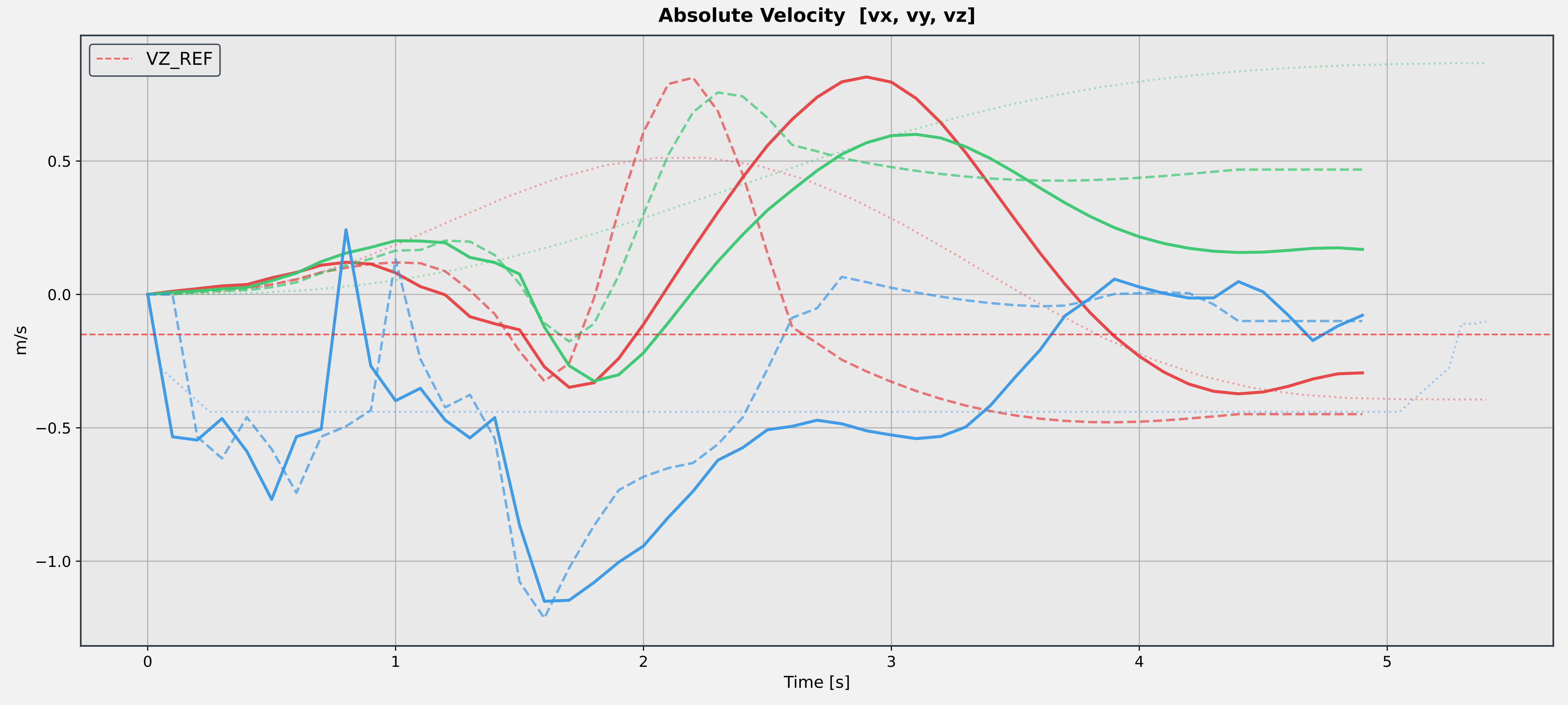
cf_1_stateswitch_mpc_alipddp_20260408_015909 — 3-D Trajectory (Absolute vs Relative)



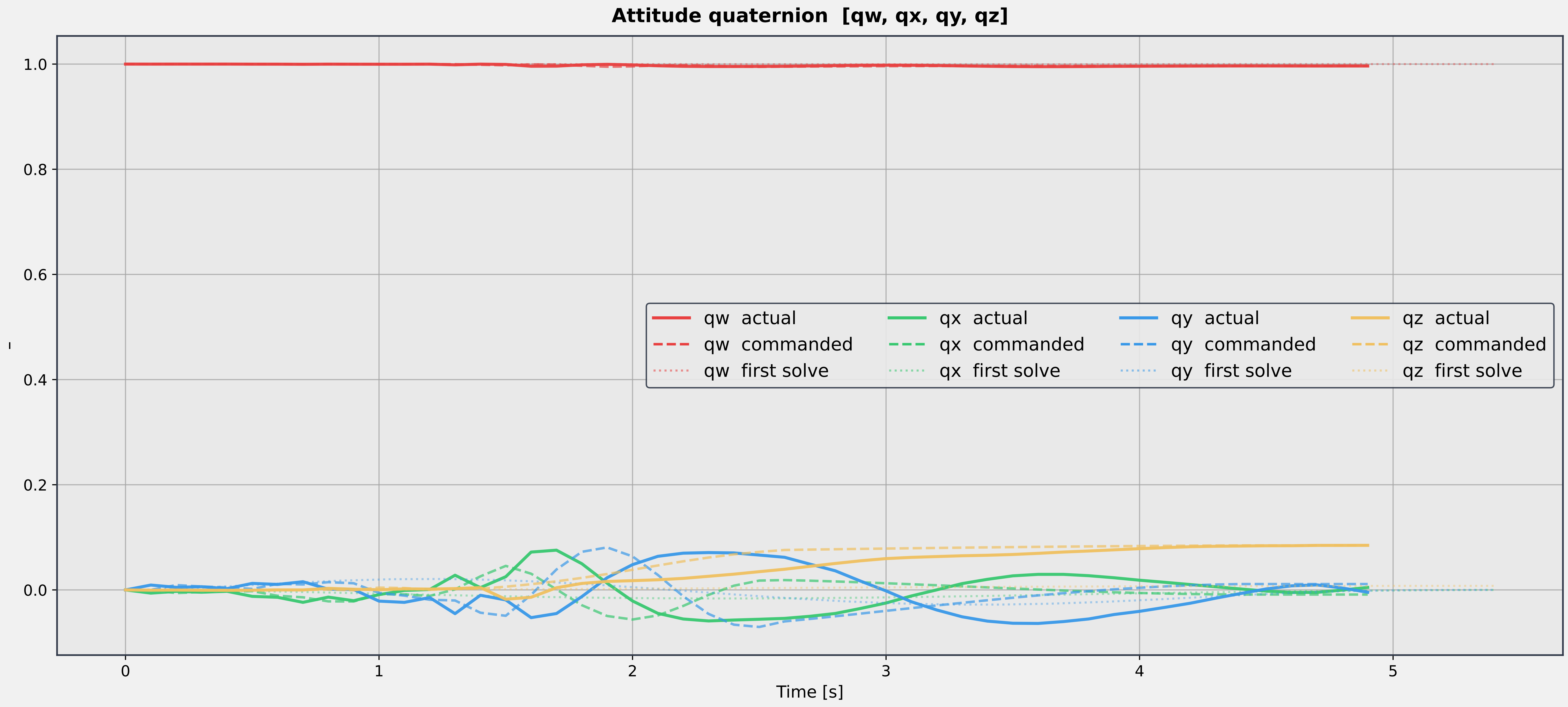
cf_1_stateswitch_mpc_alipddp_20260408_015909 — Position (Absolute)



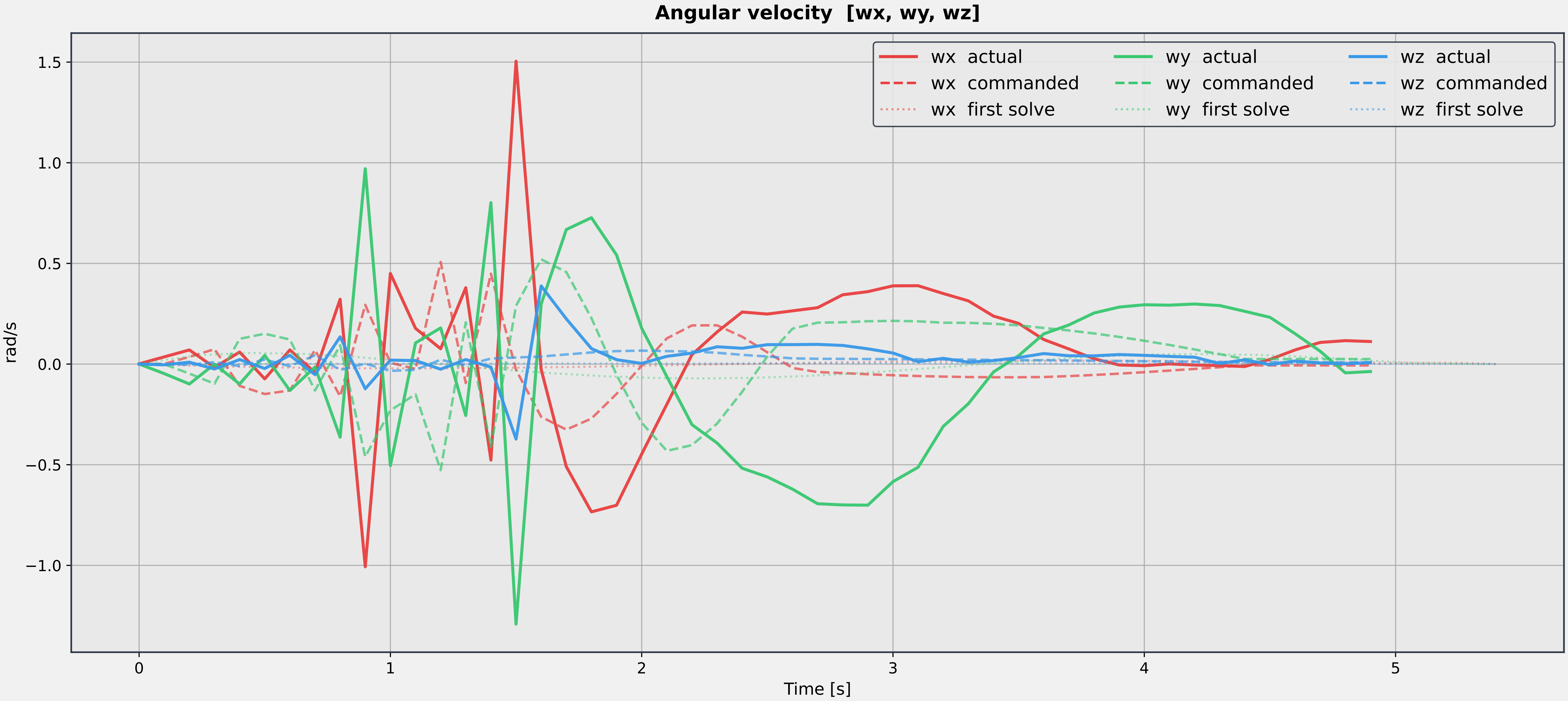
cf_1_stateswitch_mpc_alipddp_20260408_015909 — Velocity (Absolute)



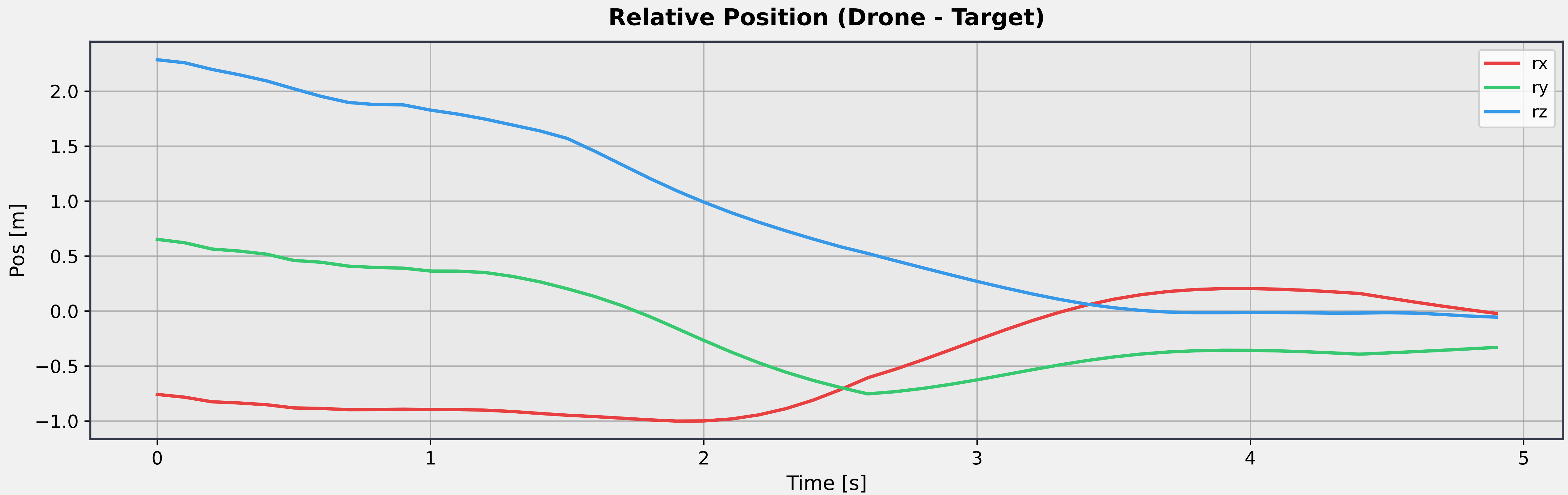
cf_1_stateswitch_mpc_alipddp_20260408_015909 — Quaternion

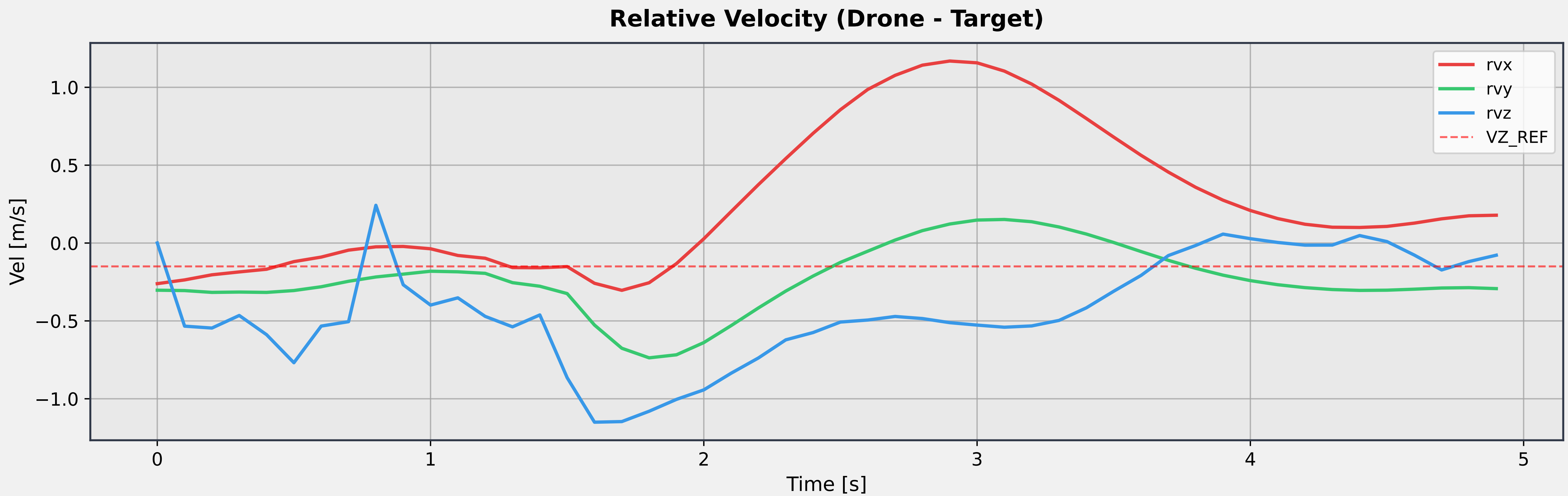


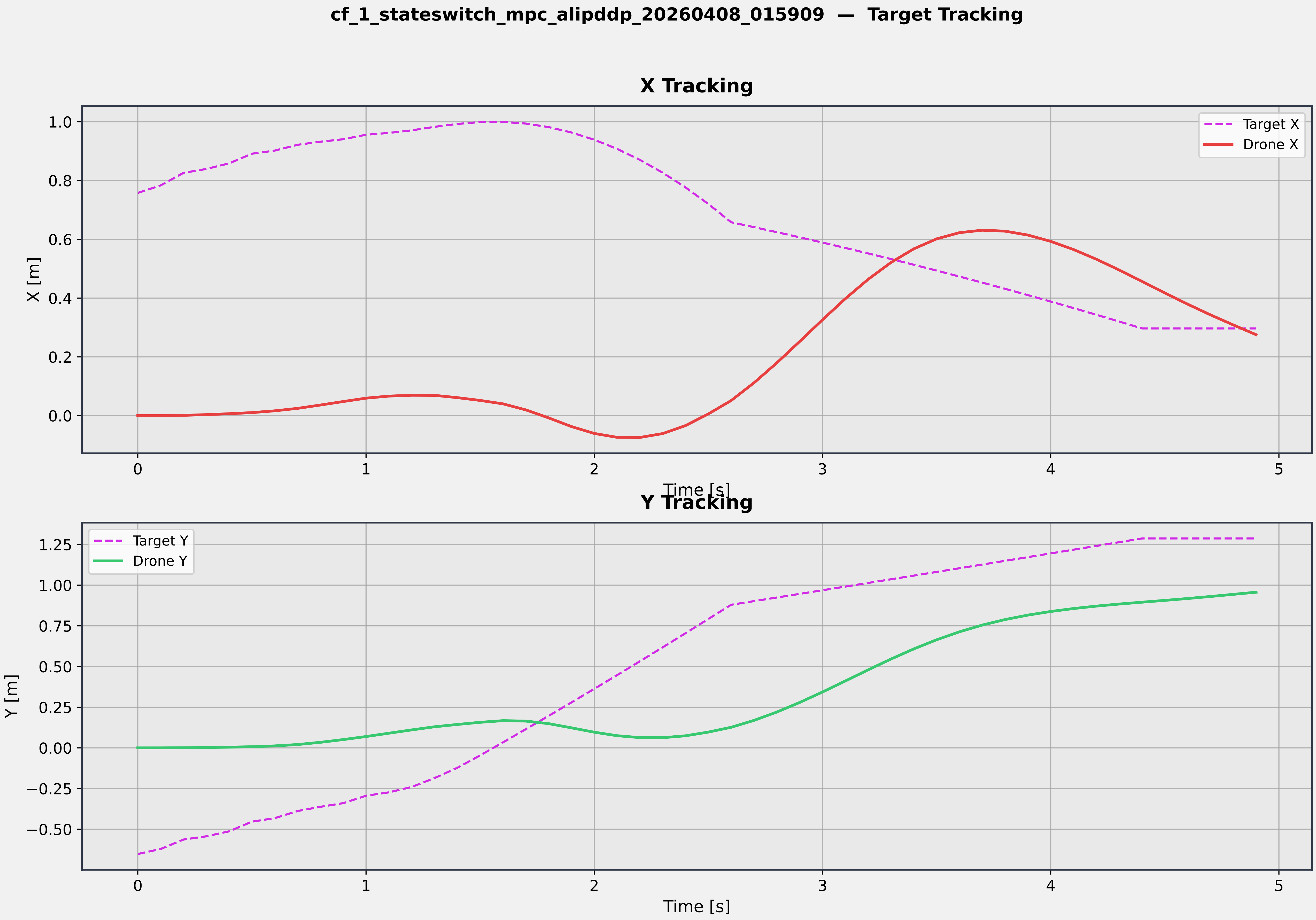
cf_1_stateswitch_mpc_alipddp_20260408_015909 — Angular Velocity



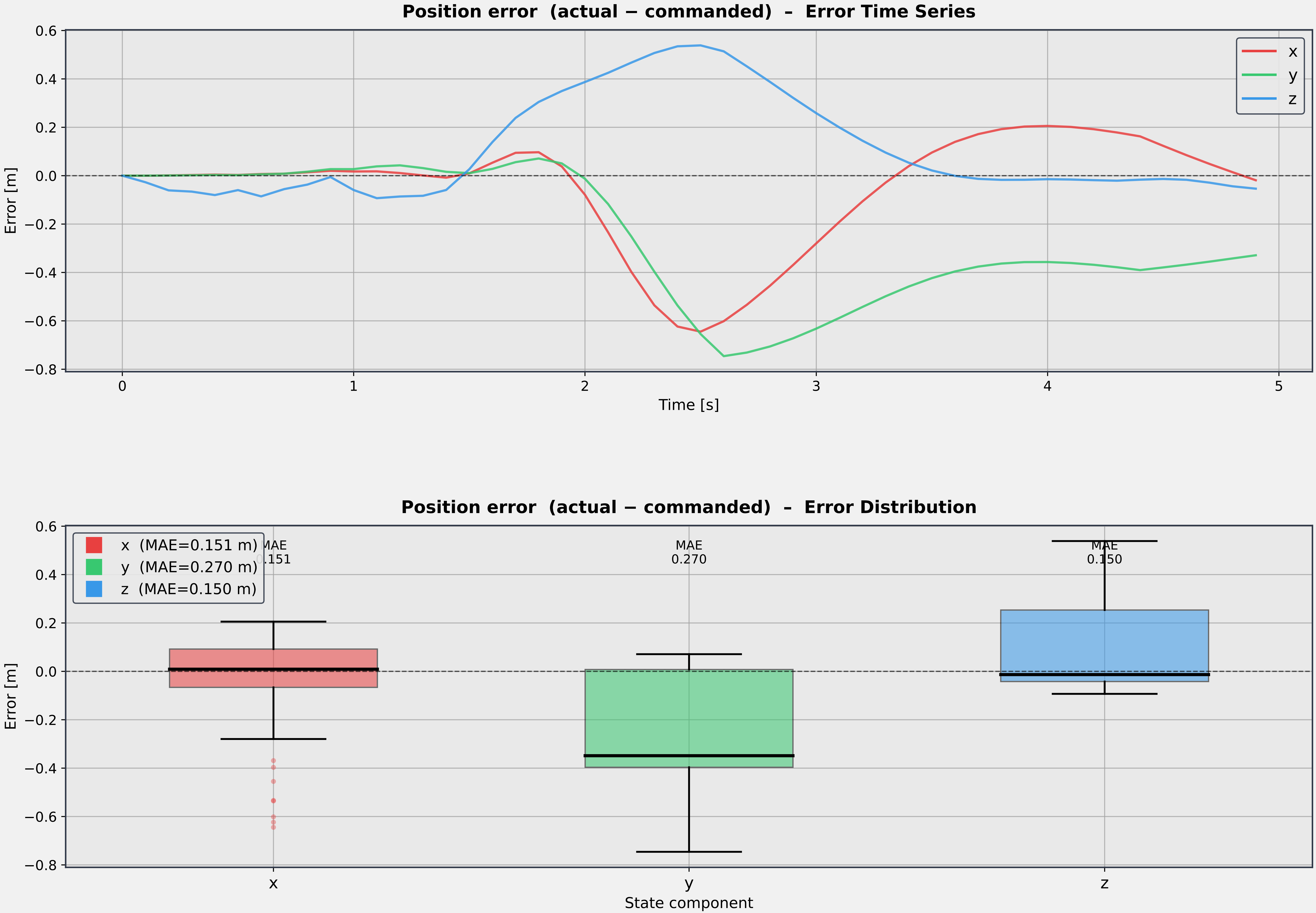
cf_1_stateswitch_mpc_alipddp_20260408_015909 — Relative States

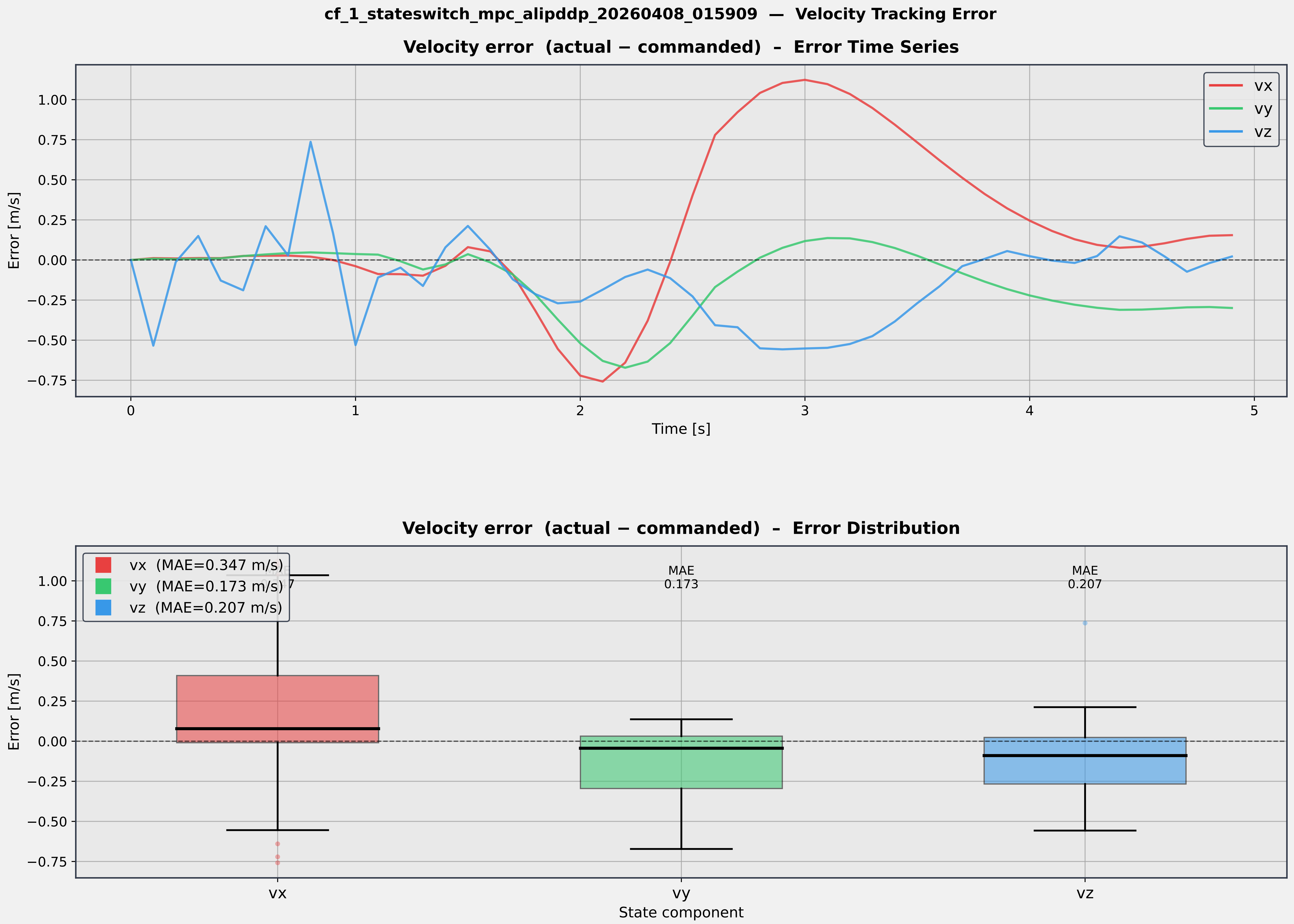




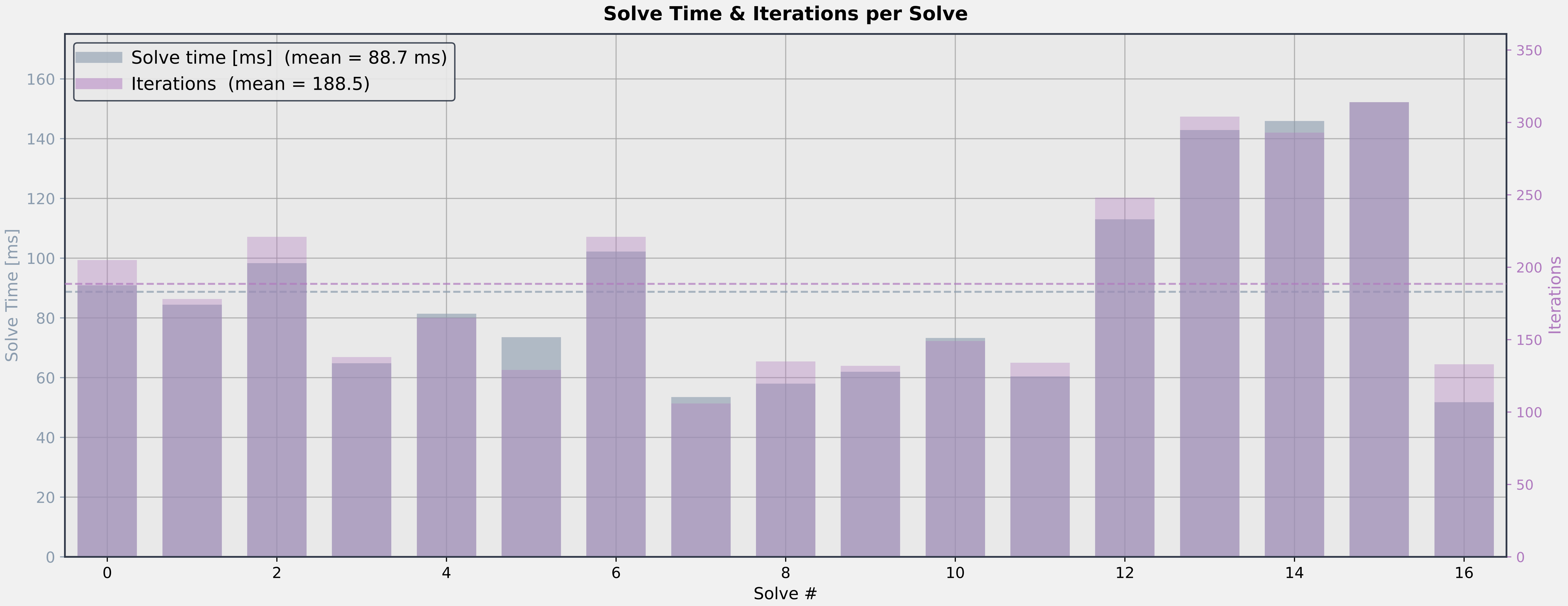

cf_1_stateswitch_mpc_alipddp_20260408_015909 — Target Tracking
X Tracking
Target X
Drone X
X [m]
Time [s]
Y Tracking
Target Y
Drone Y
Y [m]
Time [s]

cf_1_stateswitch_mpc_alipddp_20260408_015909 — Position Tracking Error




cf_1_stateswitch_mpc_alipddp_20260408_015909 — Velocity Tracking Error
Velocity error (actual − commanded) – Error Time Series
vx
vy
vz
Error [m/s]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0
1
2
3
4
5
Time [s]
Velocity error (actual − commanded) – Error Distribution
vx (MAE=0.347 m/s)
vy (MAE=0.173 m/s)
vz (MAE=0.207 m/s)
MAE
0.173
MAE
0.207
Error [m/s]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
vx
vy
vz
State component

cf_1_stateswitch_mpc_alipddp_20260408_015909 — Solver Performance



cf_1_stateswitch_mpc_alipddp_20260408_015909 — Control Inputs & Timing